

Database Relationships and Normalization Exercises

Tables and Data

Table 1: Students

StudentID	Name	Age	Major	Email
1	John Smith	20	Computer Science	john.smith@example.com
2	Jane Doe	22	Business	jane.doe@example.com
3	Ali Hassan	21	Computer Science	ali.hassan@example.com
4	Sara Lee	23	Mathematics	sara.lee@example.com

Table 2: Courses

CourseID	CourseName	Credits	InstructorID
101	Introduction to CS	3	1
102	Business Ethics	2	3
103	Calculus I	4	2
104	Algorithms	3	1

Table 3: Enrollments

EnrollmentID	StudentID	CourseID	Grade	EnrollmentDate
1	1	101	A	2024-01-15
2	2	102	B+	2024-01-16
3	1	104	A-	2024-02-01
4	3	103	B	2024-01-20
5	4	103	C+	2024-02-02

Table 4: Instructors

InstructorID	Name	Department	Email
1	Dr. Alice Johnson	Computer Science	alice.johnson@example.com
2	Dr. Bob Miller	Mathematics	bob.miller@example.com
3	Dr. Carol Davis	Business	carol.davis@example.com

Tasks for Students

1. Identify Primary Keys (PK) for each table.
2. Identify Foreign Keys (FK) and establish relationships between tables.
3. Construct Relationships:
 - One-to-Many between Students and Enrollments.
 - One-to-Many between Courses and Enrollments.

- One-to-Many between Instructors and Courses.
- Many-to-Many between Students and Courses via Enrollments.

4. Write SQL Queries:

- Find all courses taught by 'Dr. Alice Johnson'.
- List students enrolled in 'Introduction to CS'.

5. Bonus: Redesign the schema for a many-to-many relationship between Students and Courses, and explain why it's important.